

Followup 26-05

Flash pulmonary edema after a gap in heart failure medications

Patient	83-year-old female. Chronic systolic heart failure, cardiomyopathy, hypertension, type 2 diabetes, chronic kidney disease, hyperlipidemia, asthma. Prior heart failure exacerbations requiring hospitalization and mechanical ventilation. Prescribed furosemide, metoprolol succinate, losartan, spironolactone, empagliflozin, and atorvastatin. She had been unable to fill the furosemide or the metoprolol for some time because of an insurance problem , and had stayed on the empagliflozin she could still get.
Outcome	Survived. Acute hypoxic respiratory failure from flash pulmonary edema with hypertensive emergency, intubated in the field, extubated to BiPAP nine hours after hospital arrival, off BiPAP to 2 L by the following morning. No pulmonary embolism, no acute coronary syndrome. The hospital named the syndrome SCAPE in its own records , and the echocardiogram on hospital day 2 found an ejection fraction of 55% with no wall motion abnormality, unchanged from six months earlier.
Disposition	Extubated the same day and off BiPAP overnight. Diuresed to a net negative 3.3 L, with weight falling from 85 kg to 78.9 kg. Discharged on hospital day 4 to a skilled nursing facility for rehabilitation, with planned return to her daughter's home.

The call

Several days of worsening shortness of breath and swelling in her legs, and then at dawn she knocked on a family member's bedroom door and said she could not breathe and to call the ambulance. That is the whole prodrome. A heart failure patient without her diuretic fills up slowly and then crosses a line.

Crews found her sitting forward in a tripod position and folding progressively further forward while they worked. She was obtunded and not really responsive, with her entire effort directed at breathing, and she vomited shortly after arrival, which was new that morning. Lungs were wet. Pulse was strong and described as bounding. Family reported known congestive heart failure and could not give a medication list or any other history.

Field course:

- **SpO2 less than 50% on room air.** Non-rebreather at 15 L, brought to 100%.
- **Systolic 230 by palpation.** The noninvasive cuff was malfunctioning, and palpated pressures were used to guide decisions for the entire encounter.
- GCS 7, obtunded, no airway protection.
- Glucose 210.
- **ETCO2 45** by noninvasive sampling.
- Endotracheal intubation, video laryngoscopy with a bougie, 8.0 tube, grade 1 view. **The airway was noted to be swollen.** Tube at 23 cm at the teeth, secured with a commercial device, confirmed by bilateral breath sounds, absent epigastric sounds, waveform capnography, and frothy sputum in the tube.

- Bag valve mask ventilation with a **PEEP valve started at 5 and titrated to 10.**
- IV access, 22 gauge, left antecubital.
- Fentanyl 100 mcg and ketamine 200 mg for post-intubation analgesia and sedation.
- **ETCO2 into the 70s** after intubation and positive pressure.
- **Systolic 130** after intubation, sedation, and PEEP.
- No nitroglycerin given. No IV nitrate was carried, and the sublingual form is a tablet that has to dissolve under the tongue, which was full of secretions.

Weight was estimated at approximately 100 kg. She weighed 85 kg at the hospital.

Two numbers to hold: **the systolic of 230** and **the capnography going from 45 to the 70s**. Both come back later, and neither means what it looks like at first glance.

Primary impression

Acute respiratory failure with cardiogenic pulmonary edema.

Correct, and correct on the evidence available at the time. Known heart failure, wet lungs, frothy sputum in the tube, a systolic of 230, and a bounding pulse. The hospital worked the same differential with better tools and landed in the same place: CT angiography ruled out pulmonary embolism, serial troponin ruled out an occlusive coronary event, and the discharge diagnosis was acute decompensated systolic heart failure with flash pulmonary edema.

One thing to add, because it is the mechanism rather than the label. **This was not primarily a fluid problem. It was a pressure problem.** She had genuine chronic volume overload from weeks without a diuretic, and the pitting edema in her legs proves it. But the thing that put water in her alveoli between one evening and one dawn was a systolic of 230, and the treatment that fixed it was pressure applied from two directions.

Hospital course

Arrival. Intubated, gagging on the tube, tube at 23 cm at the gumline, pink tinged sputum visible inside it. Coarse breath sounds in all lobes. Two plus pitting edema in both legs. Triage vitals 160/88 with a pulse of 121. Placed on the ventilator at PEEP 8, tidal volume 450, FiO2 60%.

Chest radiograph showed diffuse prominent interstitial changes with patchy ground glass, read as pulmonary edema or pneumonia, with the tube 4.1 cm above the carina. The arterial gas came back with a pH of 7.32 and a **PO2 of 179 on 60% oxygen**, so the FiO2 was cut to 35% for hyperoxia. Propofol was started for sedation, dropped her pressure, was reduced, and the pressure climbed back.

CT angiography found no pulmonary embolism, and confirmed pulmonary edema with small bilateral pleural effusions plus bilateral multifocal ground glass and confluent consolidation in the medial lower lobes, with superimposed infection not excluded. BNP was 1,355 and the troponin rose from 19 to 46 over two hours. Lactate was 4.7 and downtrended on recheck.

Furosemide 80 mg IV went in within the first hour. Her systolic climbed back into the 200s, and a nitroglycerin infusion was started and titrated up to 90 mcg per minute over the next several hours. Spironolactone and potassium replacement were added. The nitroglycerin line infiltrated partway through and a new IV was placed.

Eight hours in, she passed a spontaneous breathing trial on pressure support of 5 over 5 at 35%. **Nine hours in**, she was extubated to BiPAP with a positive cuff leak, a strong cough, and no stridor. Nitroglycerin was stopped and nicardipine started for ongoing hypertension.

On BiPAP she began having **brief apneic pauses** and then resuming on her own. A backup rate was added. No desaturations, no dysrhythmia. The attending attributed the apneas to a central cause driven by her heart failure and noted that none had occurred during the breathing trial. Overnight she became more alert, the apneas stopped, she came off BiPAP to 2 L by nasal cannula with capnography monitoring, and the nicardipine was titrated off.

Hospital day 1. Transferred out of intensive care. The ICU summary recorded hypertensive emergency as resolved and named the presentation flash pulmonary edema and SCAPE. She had achieved roughly 3.3 L net negative on aggressive diuresis, was off BiPAP, and was answering questions appropriately.

Hospital day 2. The echocardiogram was done and it is the finding that reframes the case. Ejection fraction 55%, no regional wall motion abnormality, normal right ventricular size and function, mild mitral and tricuspid regurgitation, estimated right ventricular systolic pressure 41 mmHg, and no significant change from a study six months earlier. Intravenous furosemide was changed to oral, and guideline-directed therapy was resumed and expanded.

Days 3 and 4. Blood pressure remained the outstanding problem, running into the 170s and 180s, and spironolactone was uptitrated with a thiazide added at discharge. Physical and occupational therapy found significant debility, needing two people to assist with transfers and able to walk only a few feet with a front-wheeled walker, against a baseline of independent mobility at home. She was discharged to a skilled nursing facility for rehabilitation.

Final diagnoses: acute hypoxic respiratory failure; acute cardiogenic pulmonary edema with SCAPE; hypertensive emergency, resolved; acute kidney injury on chronic kidney disease; type 2 diabetes; elevated BNP; elevated troponin; debility.

Education points

Flash pulmonary edema is an afterload problem

Let's start with why fluid crosses into an alveolus at all. The pulmonary capillary and the alveolus are separated by a membrane thin enough for gas to cross in a fraction of a second. Whether water crosses with it depends on a balance: **hydrostatic pressure** inside the capillary pushing fluid out, against **oncotic pressure** from plasma proteins pulling it back in. Normally the pull wins and the alveolus stays dry.

Now raise the pressure. The left ventricle has to eject against whatever resistance the arterial system presents, and that resistance is **afterload**. Drive the systolic to 230 and the ventricle cannot empty properly. What it fails to eject stays behind at the end of the beat, so pressure inside the ventricle at the end of filling rises. That pressure has nowhere to go but backward, into the left atrium, into the pulmonary veins, into the pulmonary capillaries. Once capillary hydrostatic pressure exceeds what the plasma proteins can hold back, fluid crosses into the alveolus, and gas exchange stops in every alveolus it reaches.

Then the loop closes on itself. Hypoxia and the terror of not being able to breathe both dump catecholamines. Catecholamines constrict arterioles. Constricted arterioles raise afterload. Higher afterload means less forward flow and more backward pressure and more fluid in the alveoli and more hypoxia. **This is called sympathetic crashing acute pulmonary edema, or SCAPE.** It is the reason these patients go from talking to dying in under an

hour, and it is why the story here is a woman who woke up able to knock on a door and was obtunded and saturating under 50% by the time a crew reached her.

Two consequences that matter on scene. **The fluid is redistributed, not necessarily accumulated.** Many of these patients are euvoletic and some are dry, because the water was already in the body and pressure just moved it into the wrong compartment. And **the blood pressure is not a bystander, it is the disease.** A hypertensive crashing heart failure patient with a systolic in the 200s is not a patient who happens to be hypertensive. The hypertension is what is drowning them.

This patient is a hybrid, and worth saying so plainly. She had chronic volume overload from months without furosemide, which is why her legs were pitting and her BNP was 1,355. Then she had an acute afterload crisis on top of it. Both are real, they needed different treatments, and she got both.

And the echocardiogram proved the point. Her ejection fraction on day 2 was 55%, with no wall motion abnormality and no significant change from a study six months earlier. Her pump was working. She filled her own lungs with water anyway, because the problem was never the strength of the squeeze, it was the pressure the ventricle had to squeeze against and a stiff ventricle that does not tolerate a pressure spike. That combination, preserved systolic function plus impaired relaxation plus a hypertensive surge, is the classic substrate for this syndrome.

Worth knowing because it cuts against the mental model most people carry. **A normal ejection fraction does not protect you from flash pulmonary edema.** If you are waiting for a weak heart before you take a hypertensive crashing patient seriously, this is the patient who will teach you otherwise.

Further reading on SCAPE. The management side of this goes deeper than a case document can carry, particularly around nitroglycerin dosing, which is actively debated. The Internet Book of Critical Care chapter on SCAPE is free, thorough, and written at a level worth your time: <https://emcrit.org/ibcc/scape/>

PEEP is a drug, and this crew gave it

The most important therapeutic decision on this call was a PEEP valve on a bag, titrated from 5 to 10. That is not an oxygenation adjunct. In this disease it is treatment, and it works through three separate mechanisms at once.

It reopens flooded alveoli. Positive pressure at the end of expiration keeps alveoli from collapsing between breaths and pushes fluid from the air space back toward the interstitium, which restores surface area for gas exchange.

It reduces preload. Raising pressure inside the chest raises the pressure the great veins have to overcome to return blood to the heart. Less blood arrives at the right side, so less blood arrives at the congested left side.

It reduces afterload, and this is the mechanism almost nobody is taught. The left ventricle has to generate enough pressure to overcome the difference between the inside of the ventricle and the pressure surrounding it. Raise the pressure surrounding the heart, and the ventricle has less of a gradient to fight. That is why applying positive pressure to a hypertensive pulmonary edema patient does not just improve their oxygen number, it unloads the failing ventricle.

Now read her blood pressure again. **Systolic 230 before, systolic 130 after intubation, sedation, and PEEP.** Sedation contributed, and so did removing the work of breathing. But a hundred points of systolic is what afterload reduction looks like, and part of it came from a valve that costs a few dollars and clips onto the bag every truck already carries.

Prehospital considerations

Ask whether they can get their medications, not whether they are taking them. "Are you compliant with your medications" gets you a yes. "Have you been able to get your prescriptions filled" gets you the real answer. This patient had a furosemide prescription and no furosemide, and the reason was an insurance problem, not a decision. Weeks of that is the entire cause of this call, and it is the kind of detail only somebody standing in the house is positioned to find.

Positive pressure early is the treatment, not the support. CPAP or a PEEP valve on a bag is doing afterload and preload work from the moment you put it on. It is not something to try while you decide what to do.

A palpated pressure still counts. The cuff failed and the crew documented a palpated systolic of 230 and used it to drive decisions, then documented exactly why they held nitroglycerin. That is the right sequence: get a number by whatever means works, act on it, and record how you got it.

Know that a weight estimate is a guess, and that every weight-based dose inherits it. This estimate was close, at 100 kg against an actual 85 kg. Not every estimate is.

The interventions

PREHOSPITAL INTERVENTIONS

Non-rebreather at 15 L. From under 50% to 100%. That is rescue and it was right. Understand what it does and does not do here: a non-rebreather raises the oxygen concentration in whatever alveoli are still ventilating, and it does nothing for the alveoli that are full of water. The saturation came up because the working lung was given a richer mixture, not because the disease improved.

Intubation. She was obtunded at GCS 7, vomiting, and not protecting her airway. The technical execution deserves the mention: video laryngoscopy with a bougie, first look, grade 1 view, 8.0 tube, four separate methods of confirmation documented including waveform capnography, and a commercial securing device. The note that **the airway was swollen** matters, because chronic fluid overload congests the upper airway mucosa the same way it congests everything else, and an anticipated easy airway in a fluid-overloaded patient can be harder than it looks.

PEEP valve, 5 titrated to 10. Covered above. The titration is the part to notice, because it means somebody was watching the effect and adjusting rather than setting it once.

Fentanyl 100 mcg and ketamine 200 mg. Post-intubation analgesia and sedation, which is a genuine patient care issue and frequently forgotten. An intubated patient who is under-sedated is fighting the tube, driving up catecholamines and afterload in the exact patient who cannot afford it.

One note on the arithmetic. **Ketamine 200 mg against an estimated 100 kg is 2 mg per kg. Against her actual 85 kg it is 2.35 mg per kg.** Both sit comfortably inside the range for post-intubation sedation, and the estimate was a good one. It is still worth knowing that every weight-based dose inherits whatever error is in the estimate, and that the drugs where a fifteen percent difference matters are the ones with the narrowest windows.

Nitroglycerin withheld, and the reason is a route problem worth understanding. The only nitroglycerin on board was the sublingual tablet. That tablet is not swallowed and it is not absorbed through the stomach; it dissolves in saliva and the drug crosses the mucosa under the tongue directly into the venous plexus that drains there, which is how it bypasses the liver and works in a couple of minutes. Everything about that depends on a tablet sitting in contact with intact sublingual mucosa long enough to dissolve.

This patient was obtunded, had already vomited, and had significant oral secretions pooling. Put a tablet under that tongue and you cannot say where it went, how much of it dissolved, or how much crossed. You may get nothing. You may get a delayed dose from a tablet that finally dissolves after you have given up and moved on. And you have put a solid object in the mouth of a patient with no airway protection who is actively producing secretions and has already thrown up.

The blood pressure was a second reason and it is a real one, but the route was the first. Withholding it was correct here. **The general lesson is that a drug is only as good as the route it goes in by**, and sublingual absorption is the route that fails most quietly, because nothing about a tablet in a wet mouth tells you it did not work.

IN-HOSPITAL INTERVENTIONS

Ventilator at PEEP 8, tidal volume 450, FiO₂ 60%, then FiO₂ to 35%. The FiO₂ change is the one to notice. Her arterial PO₂ came back at 179 on 60%, which is far more oxygen than she needed, so it was dialed down within about twenty minutes. Rescue high, then titrate down as soon as you have a number.

Furosemide 80 mg IV. A loop diuretic, blocking sodium and chloride reabsorption in the thick ascending limb so that water follows the salt out. It treats the volume half of her problem, which was real. It does very little for the afterload half, and it does not work fast enough to be the thing that saves a crashing SCAPE patient. It is the right drug for the fluid she had accumulated over weeks.

Nitroglycerin infusion, titrated to 90 mcg per minute. Nitroglycerin is converted to nitric oxide in vascular smooth muscle, which relaxes it. At low doses the effect is mostly venous, reducing preload. **At higher doses it becomes an arterial dilator, and the arterial effect is what reduces afterload.** That distinction is why dosing in this condition is debated, and it is covered in the further reading above.

The infiltrated line. Partway through the infusion, the nitroglycerin IV infiltrated and a new one was placed. Worth flagging as a general principle: when a titrated drip stops producing the effect it was producing, check the line before you conclude the patient has changed.

Spontaneous breathing trial and extubation. Eight hours in she was placed on pressure support of 5 over 5 and tolerated it. Extubation criteria were documented properly: positive cuff leak, strong cough, adequate secretion clearance, and no stridor afterward. A **cuff leak** test is deflating the cuff to see whether air passes around the tube; if it does not, the airway around the tube is swollen and removing the tube may leave a patient who cannot breathe through their own larynx. In a patient whose airway was described as swollen on intubation, checking that is not a formality.

Extubation to BiPAP rather than to a cannula. Positive pressure did not stop being useful because the tube came out. Stepping down from invasive ventilation to noninvasive keeps the preload and afterload effects while giving the patient their airway back.

The apneas after extubation. On BiPAP she had brief apneic pauses and then resumed breathing on her own, with none during the earlier breathing trial. Heart failure produces a specific breathing pattern for a mechanical reason. The chemoreceptors that sense CO₂ and adjust breathing sit far away from the lungs, and in a failing heart the blood takes longer to travel between the two. That delay means the feedback loop is always responding to information that is out of date, so the system overshoots, then overshoots the other way, producing cycles of hyperventilation and apnea. It resolves as the heart failure resolves, and hers did overnight.

The echocardiogram. Ordered to answer the question the whole admission had been assuming an answer to. An ultrasound of the heart measures how much of the volume in the ventricle gets ejected with each beat, whether any wall segment is moving abnormally, whether the valves work, and how stiff the ventricle is during filling. Hers

came back at 55% with no wall motion abnormality, which excluded both a failing pump and a regional infarct, and left afterload and diastolic stiffness as the explanation. **It also changed her medications**, because guideline-directed therapy differs by ejection fraction, and it is the reason a heart failure patient gets one during almost every admission.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. The ranges below are the ones printed on the source reports.

Oxygenation and acid-base

Lab	Value	Reference range
pH, arterial	7.32	7.35 to 7.45
PCO2, arterial (mmHg)	46	35 to 45
PO2, arterial, on 60% oxygen (mmHg)	179	83 to 108
Bicarbonate (mmol/L)	24	22 to 29
Anion gap (mmol/L)	11	5 to 14
Base excess, arterial (mmol/L)	-2.3	-2.0 to 3.0
Potassium, point of care (mmol/L)	3.3	3.5 to 4.5
Lactic acid (mmol/L)	4.7, normal at 2 hours	0.0 to 2.0

Read the acid-base first, because it names the problem. A pH of 7.32 with a **PCO2 of 46** and a **normal bicarbonate of 24** is a respiratory acidosis. She was retaining CO2, and the reason is the same reason her capnography read in the 70s: alveoli full of fluid cannot clear carbon dioxide any better than they can take up oxygen. This is a ventilation failure caused by a cardiac problem, which is why the treatment was pressure and diuresis rather than anything aimed at her lungs.

The PO2 of 179 is the interesting one, because it is too high. She arrived saturating under 50% in her living room and an hour later had more oxygen in her arterial blood than a healthy person breathing room air. That is not a problem with the care, it is the expected result of correct rescue followed by a measurement, and the FiO2 was cut to 35% as soon as the number came back.

Divide that PO2 by the fraction of inspired oxygen and you get a **P to F ratio of about 298**, which is a real oxygenation deficit despite the impressive-looking PO2. That single division is the difference between reading a PO2 as reassuring and reading it correctly. A normal PO2 on a large amount of supplemental oxygen is a sick lung.

The capnography is the field number that needs unpacking. It read 45 before the tube and jumped into the 70s within minutes of intubation and positive pressure. She did not get worse in that interval. A noninvasive sampling line on a tachypneic patient who is tripodding and mouth breathing collects a mixture of exhaled gas and entrained room air, and room air contains almost no CO2, so the sample is diluted and the number reads low. The first measurement taken through a sealed endotracheal tube is the first true alveolar sample anybody got, and it revealed a CO2 in the 70s that had been there the whole time. She was hypoventilating badly behind a reassuring number. Treat a capnography value on a non-intubated patient as a trend to watch rather than a measurement to act on.

Lactate 4.7, and normal two hours later. Two things were making it and both resolve with the same treatment. Tissue that is not receiving enough oxygen makes lactate anaerobically, and she had been profoundly hypoxemic. Separately, catecholamines drive glycolysis through beta-2 receptors fast enough to produce lactate in tissue that is perfectly well oxygenated, and so does the enormous work of breathing she had been doing. A crashing pulmonary edema patient has all of it at once.

Notice what that lactate did not do. **Her anion gap stayed at 11 and her bicarbonate at 24**, so a lactate more than double the upper limit never opened a gap and never consumed her buffer. That fits a lactate produced fast and cleared fast rather than one accumulating against a failing circulation, and it is the reason the number was normal within two hours of nothing being aimed at it. A lactate that clears that quickly was telling you about work, not about shock.

Potassium 3.3, in a patient who has been on spironolactone and had just received 80 mg of furosemide. Loop diuretics waste potassium, which is why potassium replacement went in the same morning.

Heart failure and myocardial injury

Lab	Baseline	2 hours	Reference range
Pro B natriuretic peptide (ng/L)	1,355	not repeated	450 or less
hs troponin T (ng/L)	19	46	Less than 15 (female)
Delta from baseline (ng/L)	not applicable	27	Less than 8

BNP is released by ventricular muscle when it is stretched, which makes it a measure of wall tension rather than a measure of fluid. Hers at 1,355 against a cutoff of 450 confirms a ventricle under real strain. It rises with age and with kidney disease, both of which she has, so the number supports the diagnosis rather than making it.

The troponin is where the reading matters. It went from 19 to 46 in two hours, a delta of 27 against a threshold of 8, and the lab flagged the interpretation as abnormal. On paper that is a rising troponin with a significant delta, which is the pattern of an acute coronary syndrome.

It was not one. **This is called type 2 myocardial injury**, and it is worth separating from the type everyone is trained to look for. Type 1 is a coronary artery blocked by ruptured plaque and clot, a supply problem in one vessel. Type 2 is myocardial oxygen demand exceeding supply for reasons that have nothing to do with a blocked artery. This patient had a ventricle generating enormous pressure against a systolic of 230, a heart rate of 121, and arterial blood that had been under 50% saturated. Demand was extraordinary and supply was poor, and heart muscle leaked troponin as a result.

The direction of causation is the thing to keep. Her heart failure produced the troponin. The troponin did not produce her heart failure. She was worked up for a coronary event because that has to be excluded, and it was.

Kidney, metabolic, and the values worth reading on your own

Lab	Value	Reference range
Creatinine (mg/dL)	1.12	0.50 to 1.00
eGFR (mL/min/1.73 m2)	49	60 or greater
Blood urea nitrogen (mg/dL)	9	8 to 23

Lab	Value	Reference range
Glucose (mg/dL)	289	70 to 99
Hemoglobin A1c (%), six months prior	7.6	Less than 5.7
Albumin (g/dL)	3.2	3.5 to 5.2
Total protein (g/dL)	6.3	6.4 to 8.3
Mean cell hemoglobin (pg)	26.4	27.0 to 33.0
Mean cell hemoglobin concentration (g/dL)	29.0	32.0 to 37.0
Mean platelet volume (fL)	13.4	7.4 to 11.0

Creatinine 1.12 with an eGFR of 49 in a patient with known chronic kidney disease. A creatinine barely above the range that corresponds to a filtration rate at half of normal is worth pausing on, because **creatinine production depends on muscle mass**, and an 83-year-old woman does not produce much of it. The same filtration rate in a young muscular patient would show a much higher creatinine. The eGFR corrects for age and sex; the raw creatinine does not.

Glucose 210 in the field and 289 in the hospital, against a hemoglobin A1c of 7.6 six months earlier. That A1c says her diabetes runs moderately high on an ordinary day, so a glucose of 289 is stress hyperglycemia stacked on top of real diabetes rather than either one alone. Catecholamines drive hepatic glucose output and block insulin at the same time, and a crashing patient produces a great many catecholamines. It needed no acute treatment.

Her **blood urea nitrogen was 9**, at the bottom of the range, with a creatinine of 1.12. That ratio is low, not high, which argues against dehydration as a contributor and fits a patient who was volume overloaded rather than dry.

The albumin connects directly back to the physiology. Albumin is the main protein holding oncotic pressure inside the capillary, which is the force resisting fluid leaving for the alveolus. Hers was 3.2 against a floor of 3.5. That does not cause pulmonary edema, but it lowers the hydrostatic pressure required to produce it. Everything else being equal, she flooded at a lower pressure than a patient with a normal albumin would have.

Three values worth reading on your own. Her **mean cell hemoglobin and mean cell hemoglobin concentration were both low with a normal hemoglobin**, meaning her red cells are individually pale while she has enough of them. That is hypochromia without anemia, which is the earliest visible stage of iron deficiency or a thalassemia trait, and in a heart failure patient iron deficiency is worth knowing about because it worsens functional capacity independent of the anemia. Her **mean platelet volume was 13.4**, meaning her circulating platelets are large, which happens when the marrow is turning them over quickly. Neither needed anything done about it that day.

Learning points

1. **In crashing pulmonary edema, the blood pressure is the disease.** A systolic of 230 means the left ventricle cannot empty, which backs pressure into the pulmonary capillaries until fluid crosses into the alveoli, which causes hypoxia, which releases catecholamines, which raises the pressure further. The loop is why these patients decline in minutes. Treating the pressure treats the lungs.
2. **A PEEP valve is a cardiac drug.** It reopens flooded alveoli, reduces preload by raising intrathoracic pressure, and reduces afterload by lowering the gradient the ventricle has to eject against. This patient's systolic went from 230 to 130 across intubation, sedation, and PEEP. That is available at every level that can hold a bag.

3. **Oxygen is rescue, then it is a dose.** She went from under 50% to 100% on a non-rebreather, which was right. An hour later her arterial PO₂ was 179 on 60% oxygen and it was dialed back. Divide the PO₂ by the fraction of inspired oxygen: 179 over 0.6 is about 298, and that is a sick lung wearing a good-looking number.
4. **Capnography on a patient who is not intubated is a trend, not a measurement.** Hers read 45 before the tube and jumped into the 70s after. She did not get worse in those minutes. A tachypneic patient breathing around a sampling line entrains room air and dilutes the sample, so the number reads low; the first sealed alveolar sample tells you what the CO₂ actually was.
5. **Ask whether they can get their medications.** This entire call traces to a furosemide prescription that could not be filled because of insurance. Nothing about her adherence was the problem and asking about compliance would have found nothing. The question that finds it is whether they have been able to fill their prescriptions, and it takes about four seconds.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at <https://bcmca.github.io/followup/>